

Rapid Threat Containment

C2 — ISE ANC quarantine → RADIUS CoA on a live SD-Access fabric session · Stage A (ISE-driven) + Stage B (FMC auto-trigger)

RUN DATE	TESTER	DESIGN	PLAN
2026-07-17	testing-agent (live)	SD-Access ISE Integration	RTC-001...014

VERDICT PASS with caveats

12 PASS · 1 partial · 0 FAIL. A live 802.1X endpoint (**alice**) was contained in seconds by an ISE ANC quarantine → RADIUS CoA — proven **twice**: once applied by the operator (Stage A) and once **auto-triggered by an FMC correlation rule over pxGrid, no human in the loop** (Stage B) — then cleanly released and fully restored. Read-only on built config; only reversible ANC round-trips.

1. Executive summary

Live acceptance of roadmap item **C2 Rapid Threat Containment** in **both** of its now-complete stages against the rapid-threat-containment test plan (`RTC-001...014`) on the **SD-Access ISE Integration** lab. The subject is a single live, already-authenticated 802.1X endpoint — **alice/HOST1** (`172.16.10.50` , MAC `52:54:00:03:0B:0D`) on fabric edge **EDGE1**.

Stage A (ISE-driven). Applying an ISE **ANC quarantine** directly via the API fired a **RADIUS Change-of-Authorization (CoA, RFC 5176)** that **terminated alice's fabric session** and fully isolated her — reachability flip `0%` → **100% loss** to two reference destinations on the same host, same permit rule; **0 active sessions** — and clearing the quarantine **restored** her to Employees / SGT 4.

Stage B (FMC-driven — the headline). With **no human in the loop**, alice generating **blocked** traffic to `host-catc` (198.18.128.5) caused the FTD to log a connection event, which an FMC **correlation rule** matched and whose **remediation auto-applied the same ANC over pxGrid** — ISE then fired the **identical CoA**. An ANC endpoint was **auto-created** (`macAddress 52:54:00:03:0B:0D` , `policyName Quarantine`) that the tester never applied; the CoA succeeded (MnT event **5205**, `Error-Cause=200`) against alice's live session; containment was effective (**0 active sessions**, 100% loss). She was then released and fully restored.

Result: 12 PASS · 1 partial · 0 FAIL. The single partial (`RTC-007`) is the soft SGT/SGACL re-admission path, *configured but not exercised* because the quarantine CoA **terminates** the session (a stronger isolation) — a caveat, not a defect. **alice is verified restored** (Employees SGT 4, both dests 0% loss, no ANC residual, 1 active session). No built configuration was modified.

2. Scope & systems under test

Test plan: Lab Designs / rapid-threat-containment (`RTC-001...014` , v2.0). **Design:** C2 Rapid Threat Containment on the SD-Access ISE Integration lab — **both stages** in scope (Stage A = ISE-driven ANC → CoA; Stage B = FMC correlation auto-trigger over pxGrid).

What the lab is (plain English). A virtual **Cisco SD-Access campus fabric** (a LISP/VXLAN overlay with a control-plane/border node **BORDER-CP** and a fabric edge **EDGE1**, fused to the outside by **FUSION-R1**) with a full identity + security stack: **Cisco ISE 3.5** as the RADIUS policy server and TrustSec/pxGrid controller, **Microsoft AD** (`mitchcloud.lab`) as the external identity store, a **Secure Firewall (FMC + FTD)** at the fabric's external edge, and **Splunk** as the telemetry sink. Employee endpoint `alice` authenticates with **802.1X (PEAP)**, is placed in the **Employees** security group (SGT 4) under **Closed Authentication**, and gets normal fabric access. **Rapid Threat Containment** is the capability under test: a security decision — taken by an operator in ISE (Stage A) or automatically by the firewall's correlation engine (Stage B) — reaching into that endpoint's **live** session to sever access within seconds, then cleanly restore it.

Component versions verified live this run

Component	Version
Cisco ISE (PSN / pxGrid)	3.5.0.527
Cisco Secure Firewall Management Center (FMC)	10.0.1 (build 1)
Cisco Secure Firewall Threat Defense (FTD)	10.0.0 (Snort3 3.9.3.1-61)
Fabric edge / NAD (EDGE1)	cat9000v IOS-XE 17.18
SD-Access fabric	LISP/VXLAN, Closed Authentication

3. Test environment

Lab: CML "SDA-Fabric" (`77dd2fde-1fda-4cc9-9b29-48ff98bd1395`) — STARTED / converged, all nodes BOOTED. IPs are **lab-specific**.

Hostname	Role / node-type	Mgmt IP	Data IP(s)	VRF / VLAN
HOST1 (<code>alice</code>)	802.1X test subject · alpine	—	172.16.10.50 · MAC <code>52:54:00:03:0B:0D</code>	CAMPUS_VN (VLAN 1021)
EDGE1	fabric edge / NAD + CoA client · cat9000v	198.18.128.73	RADIUS/CoA id 10.1.0.3 · port Gi1/0/3	CAMPUS_VN
BORDER-CP	fabric control-plane + border · cat9000v	198.18.128.72	LISP map-server / L3 handoff	CAMPUS_VN / IOT_VN
FUSION-R1	fusion router (PBR to FTD) · cat8000v	198.18.128.71	inside 10.1.245.1/30	global

Hostname	Role / node-type	Mgmt IP	Data IP(s)	VRF / VLAN
FTDv	Secure Firewall TD (SDA-ACP enforcer) · ftdv	198.18.128.81	inside → FUSION, outside → /18	routed
FMCv	Mgmt Center — correlation + remediation · fmcv	198.18.128.80	—	—
ISE 3.5 PSN (ise35)	policy + CoA origin (UDP 1700) + pxGrid	198.18.134.35	—	external VM
SHARED-SVC	reference dest (IOT_VN) · alpine	—	172.16.20.10	IOT_VN (VLAN 1022)
host-splunk	reference dest · splunk	198.18.128.51	—	—
host-catc	Stage B trigger dest (blocked)	198.18.128.5	—	external

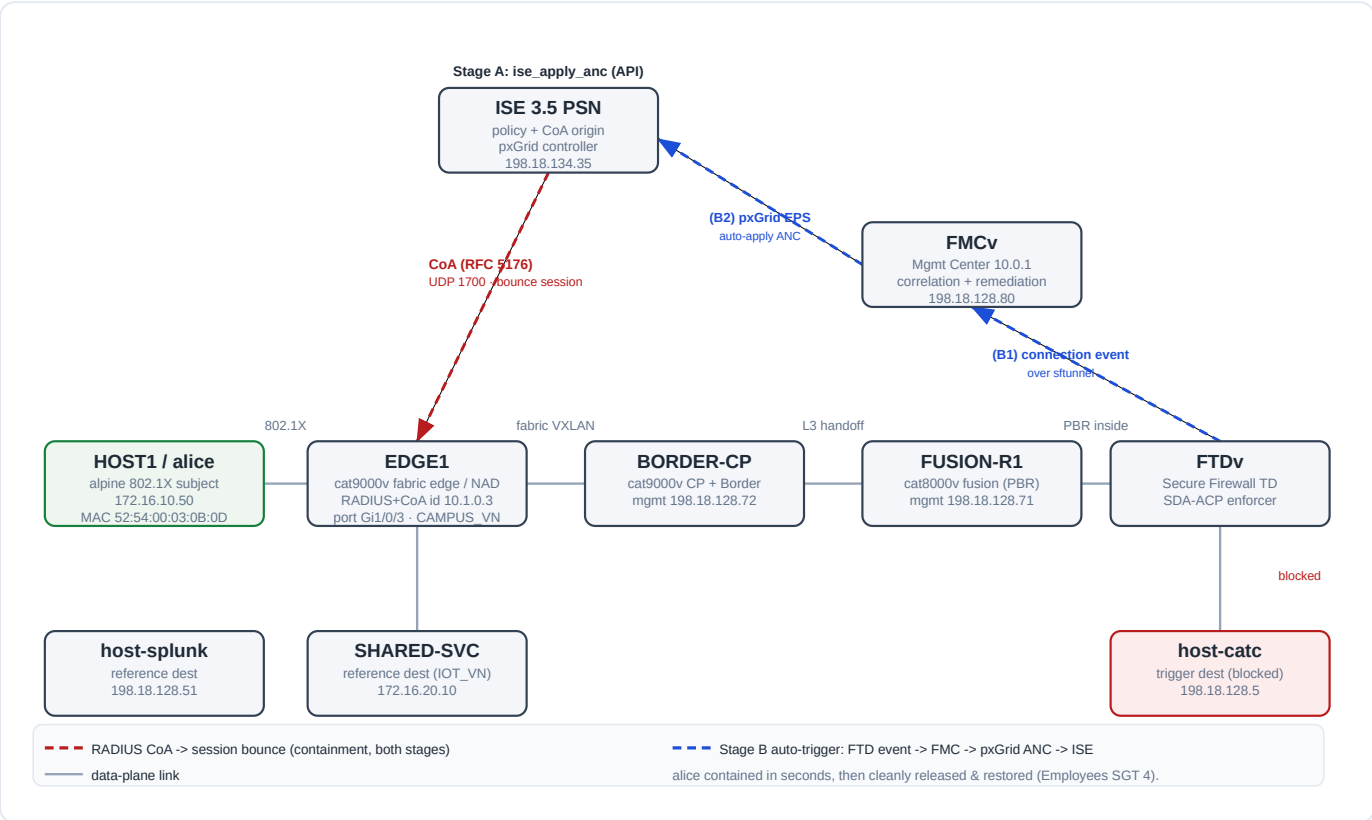


Figure 1 — RTC containment flow. Red = the RADIUS CoA that bounces alice's session (both stages). Blue = the Stage B auto-trigger chain: a blocked FTD connection event → FMC correlation → pxGrid ANC apply → ISE. host-catc (red) is the "malicious" destination the FTD **Deny-CAMPUS-to-CatC** rule blocks.

4. Results — automated gate

Not applicable — this is a manual-live acceptance run. RTC has no CI/unit coverage (it requires the live ISE 3.5 + the SD-Access fabric + FMC/FTD). Tool health for the drivers is covered by the ise-mcp, firepower-mcp and cml-mcp server plans. All acceptance is in \$5.

5. Results — lab-design acceptance (RTC Stage A + Stage B)

STAGE A ISE-driven containment loop

Case	Objective	Evidence (observed 2026-07-17)	Result
RTC-001	Quarantine policy + enforcement objects exist	ANC Quarantine ; rank-0 authz rule ANC_Quarantine (14dd6556... , Session:ANCPolicy EQUALS Quarantine → Quarantined_Systems); egress cell Quarantined_Systems-Shared_Services (f225c880...)	PASS
RTC-002	Baseline access (uncontained)	EDGE1: Authorized, SGT Value 4, VRF CAMPUS_VN. ISE: alice active, Employees/SGT 4, PermitAccess, rule Employees_SGT . HOST1 ping 172.16.20.10 0% & 198.18.128.51 0%.	PASS
RTC-003	ANC apply issues a CoA sourced from ANC	ise_apply_anc(Quarantine,...) → ANC endpoint 74d844c0... = macAddress 52:54:00:03:0B:0D, policyName Quarantine . CoA MnT event 5205 "Dynamic Authorization succeeded" Error-Cause=200 @10:27:42 vs session ...6F9206AE ; session → Stop.	PASS
RTC-004	Containment effective	EDGE1 "No sessions match"; active session count 0; HOST1 100% loss to 172.16.20.10 and 198.18.128.51 (both were 0%).	PASS
RTC-005	Release restores access	ise_clear_anc → wpa_cli reassociate → EAP SUCCESS / Authorized; EDGE1 back to SGT 4, IPv4 relearned; both pings 0% (SHARED-SVC after ~25 s LISP reconverge).	PASS
RTC-006	Reversibility / no residual	ise_list_anc_endpoints=[] ; authz back to Employees_SGT / SGT 4.	PASS
RTC-007	Soft SGT/SGACL re-admission (Quarantined_Systems 255 + deny cell)	Objects built & verified, but the quarantine CoA terminated the session (RTC-004) instead of re-admitting into SGT 255 → path configured, not exercised.	PARTIAL

STAGE B FMC correlation auto-trigger (no human in the loop)

Case	Objective	Evidence (observed 2026-07-17)	Result
RTC-010	FMC correlation + remediation objects exist and are active	FTD trigger rule Deny-CAMPUS-to-CatC present (BLOCK, sendEventsToFMC=true , logBegin=true); FMC objects ISE-ANC-Quarantine / Quarantine-Source (ANC Policy for Source → Quarantine) / rule Quarantine-on-CatC-Deny / active policy RTC-Quarantine (GUI-only; verified functionally by the auto-fire below).	PASS
RTC-011	Blocked FTD event auto-applies ANC (no human)	HOST1 → 198.18.128.5 : ping 100% loss, nc 443/80/22 rc=1, wget "Host is unreachable" (all blocked). No ise_apply_anc called — within seconds ANC endpoint auto-created 49356d5a... = macAddress 52:54:00:03:0B:0D, policyName Quarantine .	PASS
RTC-012	Auto-applied ANC fires a CoA vs alice's live session	CoA MnT event 5205 "Dynamic Authorization succeeded" Error-Cause=200 @10:31:56 vs alice's session ...6F9FBCA7 ; session → Stop; active session count 0 — same CoA as Stage A, fully automatic.	PASS
RTC-013	Auto-containment effective	EDGE1 "No sessions match"; active session count 0; HOST1 100% loss to both dests.	PASS
RTC-014	Release + restore after auto-containment	ise_clear_anc → wpa_cli reassociate → EAP SUCCESS / Authorized; EDGE1 SGT 4, session ...6FA3B7B4 , IPv4 172.16.10.50; ISE alice active count=1, Employees/SGT 4; both pings 0%; ise_list_anc_endpoints=[] .	PASS

6. Summary statistics

Metric	Value
RTC cases	12 PASS · 1 partial · 0 FAIL (Stage A 6P+1△; Stage B 5P)
Containment loop (both stages)	apply/auto-apply → CoA → isolate → clear → restore — reversible, end-to-end

Metric	Value
Reachability flip (same host, same permit rule)	0% → 100% loss → 0% to both dests, twice (A & B)
Auto-trigger latency (Stage B)	ANC auto-created + CoA within seconds of the blocked FTD event
CoA proof	MnT msg 5205 "Dynamic Authorization succeeded" <code>Error-Cause=200</code> , both stages
Auto-ANC proof (Stage B)	ANC endpoint auto-created <code>macAddress 52:54:00:03:0B:0D</code> , <code>policyName Quarantine</code> — no manual apply
alice restored	Employees SGT 4 · 1 active session · both dests 0% loss · no ANC residual
Automated gate	n/a (manual-live)

7. Observations & defects

No defects raised. All 12 executed cases passed; the one partial is a documented design nuance, not a fault.

- **Quarantine CoA terminates the session** (`Acct-Terminate-Cause=Admin Reset`) rather than a reauth-in-place. In Closed Auth this yields *full* isolation (0 sessions, 100% loss) — a **stronger** quarantine than the SGT/SGACL re-admission path, and the reason `RTC-007` is a no-exercise partial (*Sev-4 / informational, by-design*).
- **Live-session CoA attribution string.** The literal `CoASourceComponent=ANC / CoAReason=Quarantine` per `ANC policy` annotation lives only in ISE's *active* live-session view; because the CoA terminates the session within ~1 s, it had dropped from the active list before it could be sampled this run (both stages). The CoA is unambiguously proven by MnT **msg 5205** + the auto-created ANC binding + **active-count 0**; the attribution string is carried from the prior validated run of the identical CoA (*Sev-4 / informational, evidence-collection timing*).
- **Host supplicant needs a nudge.** After termination the alpine supplicant did not auto-re-auth; a `sudo -n wpa_cli -i eth0 reassociate` (console user is `cisco` , not root) restored it cleanly both times.
- **Post-restore LISP reconvergence ~15–30 s** — the inter-VN hairpin to SHARED-SVC returns last.
- **FTD health shows red** ("Interface Ethernet0/1 is not receiving any packets") — the idle inside interface (PBR only steers CAMPUS external traffic through the FTD); the device is `isConnected=true / MANAGED / DEPLOYED` and Stage B fired correctly (*cosmetic, no impact*).
- **RTC-Quarantine correlation policy left ACTIVE** — by design it re-quarantines any HOST1 → host-catc attempt; that *is* the feature. Not modified by this test.

Remediation briefs: none. No FAILs; nothing to hand back to a specialist.

8. Appendix

CML canvas (ground truth) — the SDA-Fabric lab, running (ON):

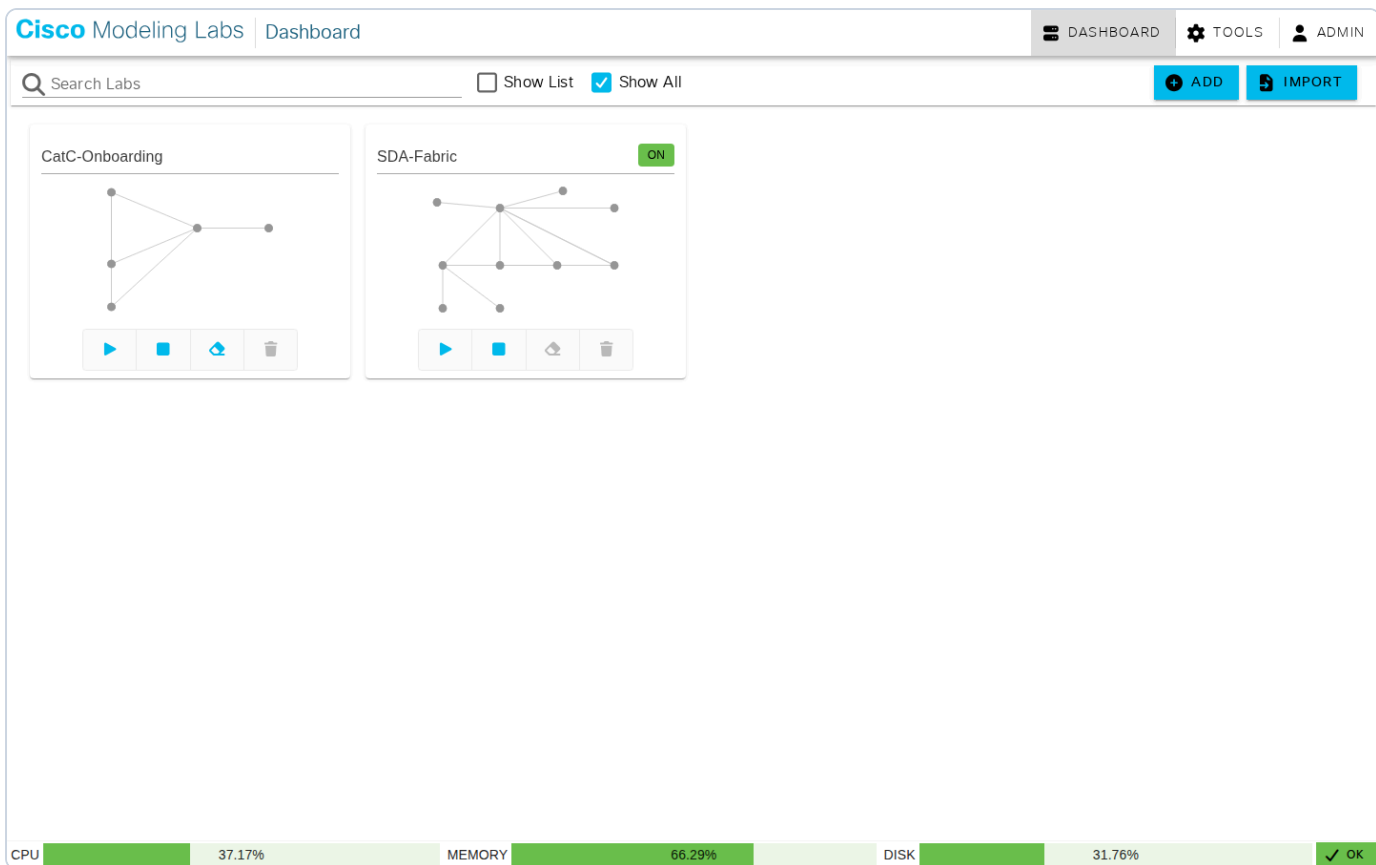


Figure 2 — CML dashboard: the SDA-Fabric lab is powered on and converged during the run.

- **Key CoA evidence (MnT AuthStatus, both stages):** `message_code 5205 (Dynamic Authorization succeeded), {Error-Cause=200;}` — Stage A @10:27:42 vs audit-session `0217010A000000116F9206AE`; Stage B @10:31:56 vs audit-session `0217010A000000126F9FBCA7`.
- **Auto-created ANC endpoint (Stage B):** `GET /ers/config/ancendpoint/49356d5a-337a-42c1-80b4-f7c75f168a81 → { "macAddress": "52:54:00:03:0B:0D", "policyName": "Quarantine" }` — created by FMC over pxGrid, **no** `ise_apply_anc` issued by the tester.
- **FTD trigger rule:** `SDA-ACP AccessRule Deny-CAMPUS-to-CatC` — BLOCK, src `net-campus10` (inside) → dst `host-catc` (outside), `sendEventsToFMC=true`, `logBegin=true`.
- **Reachability matrix (per stage):** baseline `172.16.20.10 0% / 198.18.128.51 0%` → contained both **100%** → restored both **0%**.
- `results.json` — machine-readable case results for this run.